

Visuo-motor updating in individuals with heightened autistic traits

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Abstract

Summary

Autism Spectrum Disorder (ASD) presents a range of challenges, including heightened sensory sensitivities. Here, we examine the idea that sensory overload in ASD may be linked to issues with efference copy mechanisms, which predict the sensory outcomes of self-generated actions, such as eye movements. Efference copies play a vital role in maintaining visual and motor stability. Disrupted efference copies hinder precise predictions, leading to increased reliance on actual feedback and potential distortions in perceptions across eye movements. In our first experiment, we tested how well healthy individuals with varying levels of autistic traits updated their mental map after making eye movements. We found that those with more autistic traits had difficulty using information from their eye movements to update the spatial representation of their mental map, resulting in significant errors in object localization. In the second experiment, we looked at how participants perceived an object displacement after making eye movements. Using a trans-saccadic spatial updating task, we found that those with higher autism scores exhibited a greater bias, indicating under-compensation of eye movements and a failure to maintain spatial stability during saccades. Overall, our study underscores efference copy's vital role in visuo-motor stability, aligning with Bayesian theories of autism, potentially informing interventions for improved action-perception integration in autism.

eLife assessment

This **important** study shows that a high autism quotient in neurotypical adults is associated with suboptimal motor planning and visual updating after eye movements, suggesting a disrupted efference copy mechanism. The implication is that abnormal visuomotor updating may contribute to sensory overload - a key symptom in autism spectrum disorder. The evidence presented is **convincing**, with few limitations, and should be of broad interest to neuroscientists at large.

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Introduction

Autism spectrum disorders (ASD) present a complex array of challenges, with sensory sensitivities, a recent addition to diagnostic criteria¹, standing out as a dominant aspect of this condition. Individuals with ASD often exhibit a general inflexibility in the motor domain and struggle to learn from motor errors, usually linked to restricted and repetitive behaviors^{2–4}. This difficulty in understanding motor errors can hinder their ability to adapt to new sensory information, further exacerbating sensory overload^{4–6}.

Recent research has increasingly emphasized the role of predictive abilities in ASD^{7–10}. In their daily lives within an unpredictable and often chaotic world, humans are consistently engaged in anticipating what might happen next and preparing adaptively for potential threats in their surroundings¹. Within this framework, recent theories propose that individuals with ASD may demonstrate a reduced reliance on prior information when making perceptual judgments. The Bayesian perspective suggests that the predictive system deficits in autism stem from a diminished Bayesian prior^{10,11}. From this perspective, the brain unconsciously infers information about the world by applying Bayes' rule: it assesses the probability of a prediction given sensory data, considering the likelihood of sensory data given that prediction and the Bayesian prior. The Bayesian prior is essentially a probabilistic distribution representing the expectation of the environment being in a particular state before any observations are made¹⁰. These predictive abilities include the role of perceptual priors, which play a crucial role in guiding individuals to make accurate predictions, especially when sensory data is limited. However, in cases of autism, the influence of prior knowledge is believed to be weakened, resulting in predictions that heavily rely on sensory input.

But how do these priors interact with the sensory and motor systems in individuals with ASD, and how does this influence their perceptual stability? Predicting the consequences of one's behavior is of vital importance when our movements produce changes on our sensory receptors. When we move our eyes, head, or body, we displace external stimuli on the retina. We must know about this self-produced displacement to avoid confusion with external stimulus motion¹². In other words, if we would not be able to predict the sensory consequences of our behavior, we would experience that the world is moving. The most frequent movements we perform are voluntary saccades. To rapidly direct the fovea, the area of highest resolution, towards regions of interest, we shift the eyeball with high speed about 3 times per second. Internal predictions about an imminent saccade ensure the smooth and stable continuity of our perception. This anticipation involves a signal incorporating information about the motor vector to predict the sensory consequences of the upcoming saccade. Such a signal is known as an efference copy, i.e. a copy of the motor command (for a review see¹³). During a tennis game, rapid oculomotor saccades are employed to track the high-velocity ball across the visual display. In the absence of a functional efference copy mechanism, the brain would encounter difficulty in anticipating the precise retinal location of the ball following each saccade. This could result in a transient period of visual disruption as the visual system adjusts to the new eye position. The efference copy, by predicting the forthcoming sensory consequences of the saccade, would bridge this gap and facilitate the maintenance of a continuous and accurate representation of the ball's trajectory. A key pathway responsible for transmitting efference copy signals involves the superior colliculus¹⁴, a saccade command area, to the frontal eye field via the medio-dorsal nucleus of the thalamus. Lesions affecting this pathway can impair both motor and visual updating¹³. We recently reported that in patients with lesions in thalamic nuclei, evidence for two pathways conveying efference copies exists. We found that patients with lesions in the medio-dorsal nucleus showed impairments in visual updating, while those with lesions in both the medio-dorsal and ventrolateral (VL) nuclei exhibited issues with motor updating^{15–20}.

Here, we hypothesized that the impairment of efference copy signals would result in an enduring lack of complete confirmation for pre-saccadic predictions through post-saccadic feedback. A failure in processing efference copy information might have consequences for perception around the execution of saccades. Their reliance on post-saccadic sensory feedback, rather than accurate efference copy predictions, results in a distorted perception of the visual world following eye movements (see **Figure 1** [↗](#)).

First, intra-saccadic vision might be affected. We never perceive the self-produced motion stimulation on the retina produced by our own saccades. Some theories consider saccadic omission and saccadic suppression as resulting from an active mechanism. In this view an efference copy would signal the occurrence of a saccade, yielding a transient decrease in visual sensitivity^{21 [↗](#)–23 [↗](#)}. Others however have pointed out the possibility that a purely passive mechanism suffices to induce saccadic omission^{24 [↗](#)}. A recent study has found evidence for saccadic suppression already in the retina. Idrees et al. ^{25 [↗](#)} demonstrated that retinal ganglion cells in isolated retinæ of mice and pigs respond to saccade-like displacements, leading to the suppression of responses to additional flashed visual stimuli through visually triggered retinal-circuit mechanisms. Importantly, their findings suggest that perisaccadic modulations of contrast sensitivity may have a purely visual origin, challenging the need for an efference copy in the early stages of saccadic suppression. However, the suppression they measured lasted much longer than time-courses observed in behavioral data. An efference copy signal could thus be necessary to release perception from the retinal suppression.

Second, the internal updating of the spatial representation for the saccade vector would be affected. Since intra-saccadic vision is omitted, visual perception is confronted with a retinal image before the saccade and a different one after the performance of the saccade. Predicting the appearance of the post-saccadic image leads to a remapping of internal space and thereby to a seamless perception across the saccade^{26 [↗](#)}. If the remapping process is not accurate, the appearance of the post-saccadic image would come as a surprise. Surprising stimuli appear much more salient than predicted ones. Failures in trans-saccadic remapping in individuals with autistic symptomatology could be the reason or at least contribute to the sensory overload they experience.

Third, in addition to problems in visual updating, a lack of proper efference copy transfer should produce problems in motor updating. In cases in which visual information is not available or reliable, the sensorimotor system must rely on internal estimates about its own movements.

In the laboratory, trans-saccadic updating can be investigated in motor behavior and in perceptual localization. In the double-step paradigm, two consecutive saccades are made to briefly displayed targets^{27 [↗](#),28 [↗](#)}. The first saccade occurs without visual references, relying on internal updating to determine the eye's position. The accuracy of the second saccade quantifies the first saccade's updating process. Perceptual updating is tested similarly, with subjects remembering the position of a target and then indicating it relative to a second target after a single saccade. In this paradigm, one stimulus is shown before and another after saccade execution. Together these two stimuli produce the perception of “apparent motion”. If stimuli are placed such that the apparent motion path is orthogonal to the saccade path, then the orientation of the apparent motion path indicates how the saccade vector is integrated into vision. The apparent motion trajectory can only appear vertical if the movement of the eyes is perfectly accounted for, that is the retinotopic displacement is largely compensated, ensuring spatial stability. However, small biases of motion direction – implying under-(or over-) compensation of the eye movement – can indicate relative failures in this stabilization process. In a seminal study, Szinte and Cavanagh^{29 [↗](#)} found a slight over-compensation of the saccade vector leading to apparent motion slightly tilted against the direction of the saccade. More importantly, when efference copies are not available, i.e. localization occurring at the time of a second saccade in a double step task, a strong saccade under-

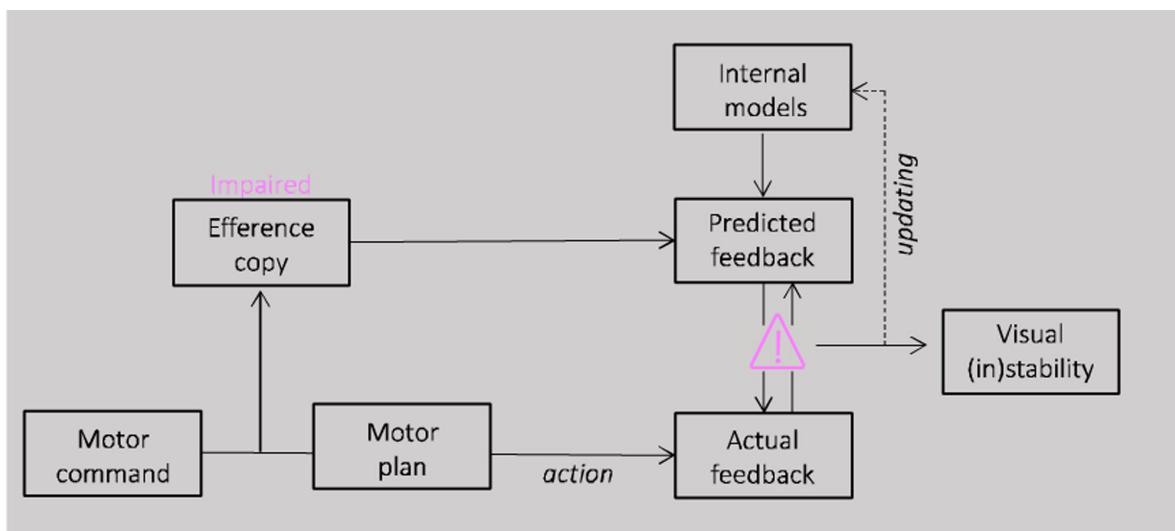


Figure 1.

Schematic illustration of the proposed model. For saccade generation, a motor command is sent to the motor plan which produces the desired eye movement. A correct plan of the movement depends on a comparison between the position of the eye before and after a saccade and therefore on the accuracy of the efference copy signal. If the efference copy is impaired, pre saccadic prediction will be never completely confirmed by the post saccadic feedback. This constant lack of successful match between prediction and sensory evidence could influence the accumulation of sensory predictions over time, leading to a less stable perception.

compensation occurs³⁰. This phenomenon cannot be explained by perisaccadic mislocalization of flashed visual stimuli^{31,32}, but the two phenomena may be related in that they may both depend upon efference copy information.

In the present study, we directly investigated whether the precision of efference copy signals influences the accumulation of sensory predictions over time and explored potential variations in this mechanism among individuals with a heightened autistic trait. The autistic traits of the whole population form a continuum, with ASD diagnosis usually situated on the high end^{33–35}. Moreover, autistic traits share a genetic and biological etiology with ASD³⁶. Thus, quantifying autistic-trait-related differences in healthy people can provide unique perspectives as well as a useful surrogate for understanding the symptoms of ASD^{33,37}. In two experiments, we focused on the role of efference copy in the pathophysiology of autistic traits, particularly in maintaining perceptual continuity during saccades. We hypothesized that individuals with a high autistic phenotype would exhibit decreased accuracy in second saccades during double-step tasks and under-compensation when localizing a stimulus based on efference copy signals. These findings could shed light on the visual stability and predictive challenges faced by individuals with ASD.

Results

Motor updating

We investigated the influence of autistic traits on visual updating during saccadic eye movements using a classic double-step saccade task. This task relies on participants making two consecutive saccades to briefly presented targets. The accuracy of the second saccade serves as an indirect measure of how effectively the participant's brain integrated the execution of the first saccade into their internal representation of visual space. Participants were divided into quartiles based on the severity of their autistic traits, as assessed by the Autistic quotient questionnaire (cite). We hypothesized that individuals with higher autistic traits would exhibit greater difficulty in visual updating compared to those with lower autistic traits. This would be reflected in reduced accuracy of their second saccades in the double-step task. **Figure 2C** illustrates examples from participants at the extremes of the autistic trait distribution (Autistic quotient = 3, in orange and Autistic quotient = 31, in magenta). As shown, both participants were instructed to make saccades to the locations indicated by two brief target appearances (T1 and T2), as quickly and accurately as possible, following the order of presentation. However, successful execution of the second saccade requires accurate internal compensation for the first saccade, without any visual references or feedback available during the saccade itself.

To control for general impairments in saccade generation, we first checked for the accuracy of the first saccade (from FP to T1). **Figure 3A** shows the average saccade landing positions of the two subgroups of participants (lower quartile and upper quartile of the AQ distribution in orange and magenta respectively) in the double-step task. For the analysis we collapsed data from trials with second saccades into the lower and upper visual field. For statistical analyses, we first compared horizontal landing positions of the first saccade for the two groups (**Figure 3B**). Saccade landing positions of participants in the lower quartile (mean degree $\pm$ SEM: 10.17 ± 0.50) did not deviate significantly from those in the upper quartile (mean degree $\pm$ SEM: 9.65 ± 0.77). This result was also confirmed by a two sample t-test ($t(14) = 0.66$; $p = 0.66$, $BF_{10} = 0.40$) and by a non-significant correlation between the landing position of the first saccade and the autistic traits of our participants ($r = -0.09$; $p = 0.55$, $BF_{10} = 0.14$, not shown as a figure).

We then calculated the angle between the participants' second saccade vector and the optimal vector that would have led to the physical target position T2 from the second saccades' starting position. The analysis of the second saccade provided the means to determine the effect of

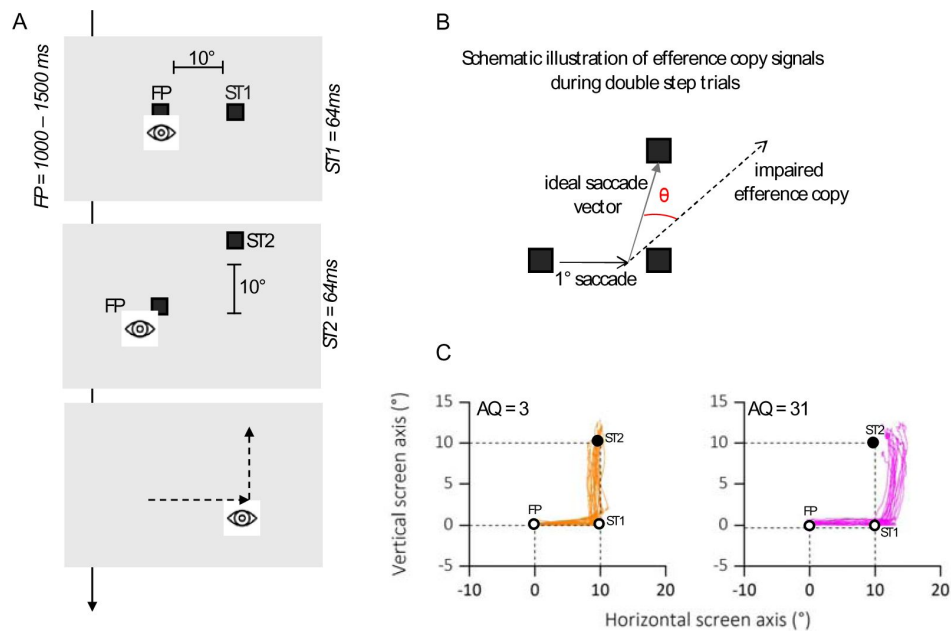


Figure 2.

A) Time course of presentations and spatial arrangement of stimuli during double step task. Participants made double-step saccades from a fixation point (FP) to the first saccadic target (ST1), and from there to a second saccade target (ST2). Both FP and saccade targets were separated by 10°. Participants were instructed to initiate the saccade only after all stimuli had disappeared, so to avoid any visual references. **B)** Hypothetical pattern of first and second saccades on a particular trial and illustration of the variables angle shift and horizontal endpoint shift of second saccades. We determined how far the second saccade deviated from the optimal vector that would have directed the gaze onto the target. We calculated the angle between the second saccade vector and the optimal vector connecting the starting position of the second saccade and the second saccade target. These angles were computed separately for upward and downward saccades, but since they resulted in very similar values, we collapsed the data into upward angles only. **C)** Eye position traces for two representative participants. The eye position between the initiation of the first saccade from FP toward ST1 (empty circles) and the termination of the second saccade toward ST2 (filled circle) are depicted for each trial in which successive saccades to ST1 and ST2 were produced. Plots are separated for low autistic traits (AQ = 3, in orange) and high autistic traits (AQ = 31, in magenta) example participants.

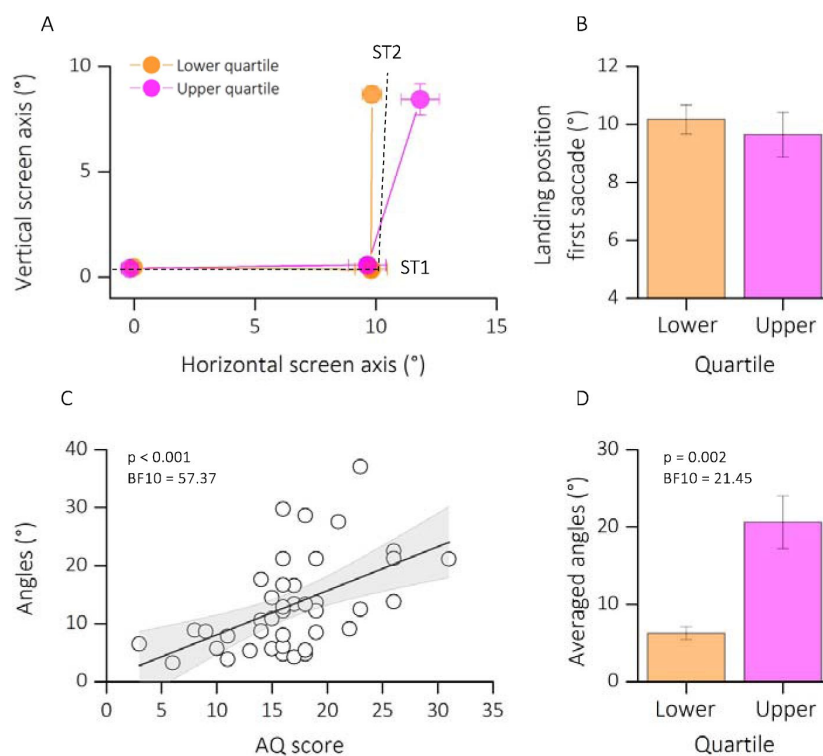


Figure 3.

A) Average saccade vectors from FP (0, 0) to ST1 (10, 0) and from ST1 to ST2 (10, 10). Data are shown for a subsample of participants, falling in the lower (orange) and upper (magenta) quartile of the autistic quotient distribution. Data from upward and downward second saccades are collapsed. Dashed lines indicate the required saccade path from FP to ST1 and from ST1 to ST2. **B)** Average first saccade landing position for lower and upper quartile of the AQ distribution. Error bars are SEM. **C)** Linear regression between angles (distance the actual second saccade vector and the optimal vector that would have led to the physical target position ST2 from the second saccades' starting position) and the Autistic Score of participants. Text insets report p-values and associated Bayes Factors of Pearson's Rho. Thick black line shows the linear fit through the data. The shaded area represents the standard error for each observation mean as predicted by the regression line. **D)** Average angles (°) for lower (AQ < 13, orange) and upper (AQ > 19, magenta) quartile of the AQ distribution. Error bars are SEM. Text insets report p-values and associated Bayes Factors of paired sample T-test.

effeference copies signals in underlying extraretinal information. If the participant fails to compensate for the first saccade due to disrupted effeference copy, the second saccade vector deviates from the location of the second target (**Fig. 1B** [↗](#)). A correct remapping of the previously seen target position would lead to smaller angle of deviations from the optimal saccade vector; however, a failure in using effeference copy-based updating of visual space would results in larger deviations. **Figure 3C** [↗](#) plots the motor bias (expressed as angle of deviations) as a function of autistic traits severity. We found a significant correlation between the autistic characteristics of our group of participants and the amount of motor bias ($r = 0.52$; $p < 0.001$; $BF_{10} = 57.37$). This result is also confirmed by a strong group difference, between the lower and upper quartile of AQ distribution ($t(14) = -4.68$; $p = 0.002$; $BF_{10} = 21.45$), with a mean bias of 6.26 ± 0.80 for the lower quartile and of 20.61 ± 3.40 for the upper quartile.

Results on the double step task suggest that although participants did not show an impairment in the first saccade execution, those with higher autistic traits reported difficulties in using extra-retinal information about the amplitude and direction of the motor vector of the first saccade, in order to update the spatial representation of the second target. Thus, they were less able to use the effeference copy of the first saccade to construct a spatial representation of the second target location in generating an accurate saccade towards it.

Visual updating

With a trans-saccadic localization task, we explored how autistic traits affect the integration of eye movements into visual perception. Participants were presented with stimuli before and after a single saccade, creating an illusion of apparent motion. We measured the perceived direction of this displacement, which is influenced by how well the participant's brain accounts for the saccadic eye movement. We predicted that individuals with higher autistic traits would show a stronger bias in the perceived displacement direction, suggesting a less accurate integration of the eye movement into their visual perception.

Participants were instructed to report the position of a second dot relative to a first dot, after saccade execution (see **Figure 4A** [↗](#)). The two dot stimuli, presented below and above the saccade path, were horizontally displaced from each other to a variable degree. One dot was always presented before saccade onset, the other after saccade completion, so that the saccade dissociates retinal from screen coordinates. To perceive vertical trajectory in screen coordinates, the saccade vector must be compensated for – likely through effeference copy information. At saccade completion, on the retina the first probe falls to the right side of the fovea while the second falls to the left side (see **Figure 4B** [↗](#), retinal coordinates). To compensate for the effects of the saccade, the visual system corrects the expected location of T2 in the opposite direction to the saccade (black arrow, remapped target position). If this correction is accurate, the displacement of the dots is perceived relatively vertical (**Figure 4B** [↗](#), screen coordinates), rather than in retinotopic direction (relatively oblique, gray dashed line) and space constancy is maintained. Every shift from the vertical trajectory indicates an under or over-compensation of the saccade vector. Previous studies have shown that the dots trajectory is indeed seen as approximately vertical, although a systematic slant of trajectories implies a tendency to overcompensate for the saccade vector^{29,38} [↗](#). However, if the retinotopic displacement is uncompensated, a failure in spatial stability is encountered. **Figure 5A** [↗](#) plots two example psychometric functions for one participant at the extremes of the lower (AQ = 3, in orange) and upper quartile of the distribution (AQ = 31, in magenta). The mean of the psychometric function (PSE) represented the location at which the references and the probe appeared to be vertically aligned. The PSE for the two example participants resulted very different: while $\sim 0^\circ$ bias represents accurate, spatiotopic correction, a negative bias (PSE < 0) reflects an under compensation of the saccade vector.

We first checked that all participants were able to execute accurate saccades. The saccade amplitude did not correlate with the autistic traits of our participants ($r = -0.18$; $p = 0.33$; $BF_{10} = 0.22$, not shown as a figure). Moreover, when checking on the mean amplitude across trials (**Figure**

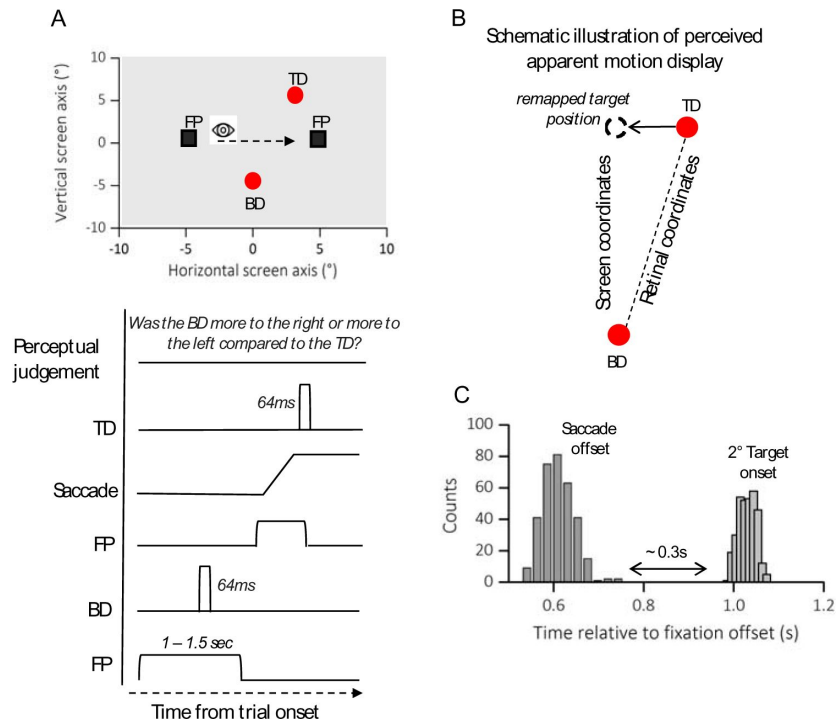


Figure 4.

A) Spatial arrangement of stimuli and time course of presentations during trans-saccadic localization task. Participants made a saccade from a fixation point (FP, left) to a second fixation point (FP, right). At a random time before the first FP disappearance, a red circle (Bottom Dot, BD) was presented for 64ms, at saccade completion a second red dot (Top Dot, TD) was delivered for another 64ms. Participants were instructed to initiate the saccade only after the first FP had disappeared. At saccade completion participants were asked to report whether the first dot delivered (in the figure Bottom Dot, BD) was more to the right or more to the left compared to the second one (in the figure Top Dot, TD). **B)** Schematic illustration of perceived apparent motion display. On the retina, the first probe falls to the right side of the fovea while the second falls to the left side. To compensate for these effects of the saccade, the visual system corrects the expected location of the second dot in the opposite direction to the saccade (remapped target position, gray dot). If this correction is accurate, the displacement is perceived in its spatiotopic (vertical dashed line: from remapped target position to BD) rather than retinotopic arrangement (oblique dashed line: from TD to BD) and space constancy is maintained. **C)** Average frequency distribution of saccade offsets and second target onset. We analyzed only the trials in which the second target was delivered after saccade completion (on average around 0.3 sec), so that the visual system could dissociate the retinal from the screen coordinates.

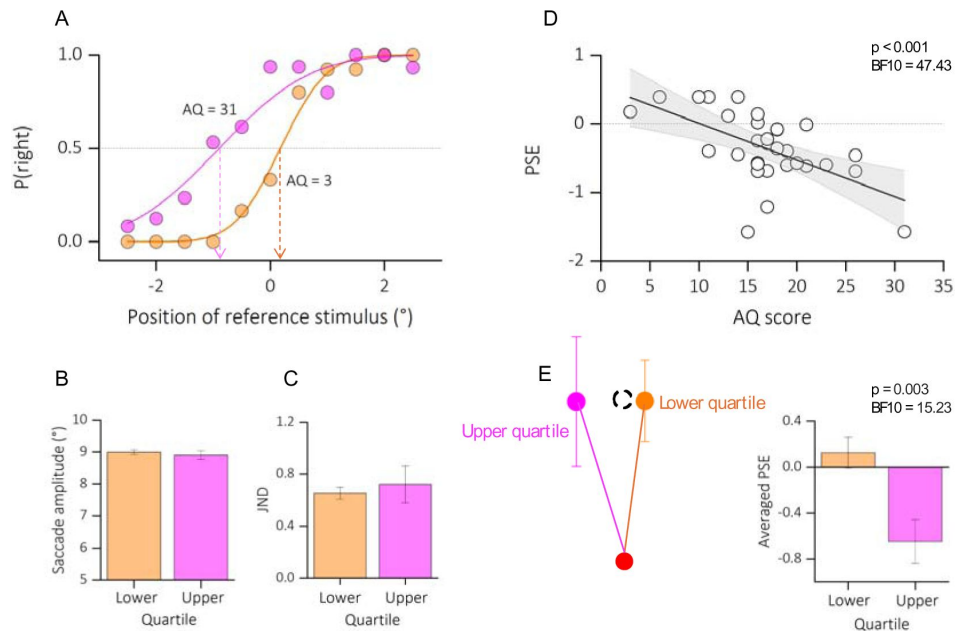


Figure 5.

A) Psychometric functions for two example participants at the very extremes of the autistic quotient distribution (AQ = 3, in orange; AQ = 31, in magenta). The functions plot the proportion of reporting "right", as function of the position of the second dot relative to the first one (shown in the abscissa). The vertical-colored lines show the estimates of the PSE, given by the median of the fitted cumulative Gaussian functions. **B)** Average saccade amplitude (°) for lower (AQ < 14, in orange) and upper (AQ > 20, in magenta) quartile of the autism distribution. Saccade amplitude did not vary with autistic traits. **C)** Precision thresholds (JND) for lower and upper quartile of the AQ distribution, same convention as in B. Here, the precision in reporting the dot displacement was similar between sub-samples. **D)** Linear regression between PSE and the Autistic score of participants. Text insets report p-values and associated Bayes Factors of Pearson's Rho. Thick black line shows the linear fit through the data. The shaded area represents the standard error for each observation mean as predicted by the regression line. **E)** Schematic illustration of perceived dots displacement for upper and lower quartile of the autistic quotient. A localization compensating fully for the saccadic eye movement would lead to a perfect vertical localization (gray dot, dashed line). However, deviations from the vertical indicate and under or overcompensation of the saccade vector. We found a large under compensation of the saccade vector for the participants with higher autistic traits, compared to participants with lower autistic traits. See also bar plot, average PSE, on the right. Text insets report p-values and associated Bayes Factors of two sample T-test.

5B [↗](#)), we did not find a significant difference between the lower (mean degree $\pm$ SEM: 8.66 ± 0.27) and upper (mean degree $\pm$ SEM: 7.96 ± 0.46) quartile of participants AQ distribution (two sample t-test, $t(13) = 1.42$; $p = 0.17$; $BF_{10} = 0.62$).

We then checked that all participants were able to complete the task. A measure of precision (JND) was extracted from each participant's psychometric function. **Figure 5C** [↗](#) shows the averaged JND for lower and upper quartile of the distribution. We report a non-significant difference between the upper and lower quartile of the autistic score's distribution ($t(13) = -0.46$; $p = 0.64$; $BF_{10} = 0.29$) nor a significant correlation between JND and the autistic quotient of all our participants ($r = 0.17$; $p = 0.36$; $BF_{10} = 0.21$, not shown as a figure). Indeed, the averaged precision in reporting the position of the second dot was very similar between the upper (mean $\pm$ SEM: 0.72 ± 0.04) and lower (mean $\pm$ SEM: 0.65 ± 0.14) quartile of the distribution.

By contrast, the averaged biases (expressed as the PSE) varied as a function of the autistic trait severity (**Figure 5D** [↗](#)): the higher the autistic scores the stronger the bias in reporting the displacement in retinotopic coordinates, hence correcting less for the saccade eye movement ($r = -0.58$; $p < 0.001$; $BF_{10} = 47.43$).

Indeed, while we report a slightly tilt of the dots trajectory for the lower quartile of the distribution (mean $\pm$ SEM: 0.12 ± 0.13), the reported bias was tilted in the opposite direction for participants falling in the upper quartile (mean $\pm$ SEM: -0.64 ± 0.19), implying an under-compensation of the saccade vector and a partial failure in visual stability (see **Figure 5E** [↗](#) for a schematic illustration of the results). This result is also confirmed by a significant difference between the two groups ($t(13) = 3.63$; $p = 0.003$; $BF_{10} = 15.23$; Fig4E, bar plot).

We finally examined the relationship between the motor and visual biases in the participants that completed both tasks. **Figure 6** [↗](#) plots the angles of deviations (motor bias) as a function of the PSE of our participants (visual bias). We found that motor and visual biases very strongly correlated ($r = -0.57$; $p < 0.001$; $BF_{10} = 37.4$), suggesting that a similar mechanism – controlled by the same efference copy signal – might be responsible for both motor and visual updating.

Understanding how these biases might change over time could provide further insights into this mechanism. Specifically, we investigated whether participants exhibited any learning effects throughout the experiments. For data of Experiment 1 – motor updating – we divided our data into 10 separate bins of 30 trials each. We conducted a repeated measure ANOVA with the within-subject factor “number of sessions” (two main sessions of 5 bins each, ~150 trials) and the between-subject factor “group” (lower vs upper quartile of the AQ distribution). We found no main effect of “number of sessions” ($F(1,7) = 0.25$, $p = 0.66$), a main effect of “group” ($F(1,7) = 2.52$, $p = 0.015$), and no interaction between the two subsample of participants and the sessions tested ($F(1,7) = 0.51$, $p = 0.49$). Data of Experiment 2 – visual updating – were separated into 3 sessions. For each session we extracted the PSE and we conducted a repeated measure ANOVA with within subject factor “sessions” and between subject factor “groups” (lower vs upper quartile of the AQ distribution). Also here we found no main effect of sessions ($F(1,13) = 0.86$, $p = 0.39$), a main effect of group ($F(1,14) = 11.85$, $p = 0.004$), and no interaction between the two subsample of participants and the sessions tested ($F(1,13) = 0.20$, $p = 0.73$). In conclusion, the current study found no evidence of learning effects across the experimental sessions. However, a significant main effect of group was observed in both Experiment 1 (motor updating) and Experiment 2 (visual updating). Participants in the group with higher autistic traits performed systematically differently on the task, regardless of the number of sessions completed compared to those in the group with lower autistic traits.

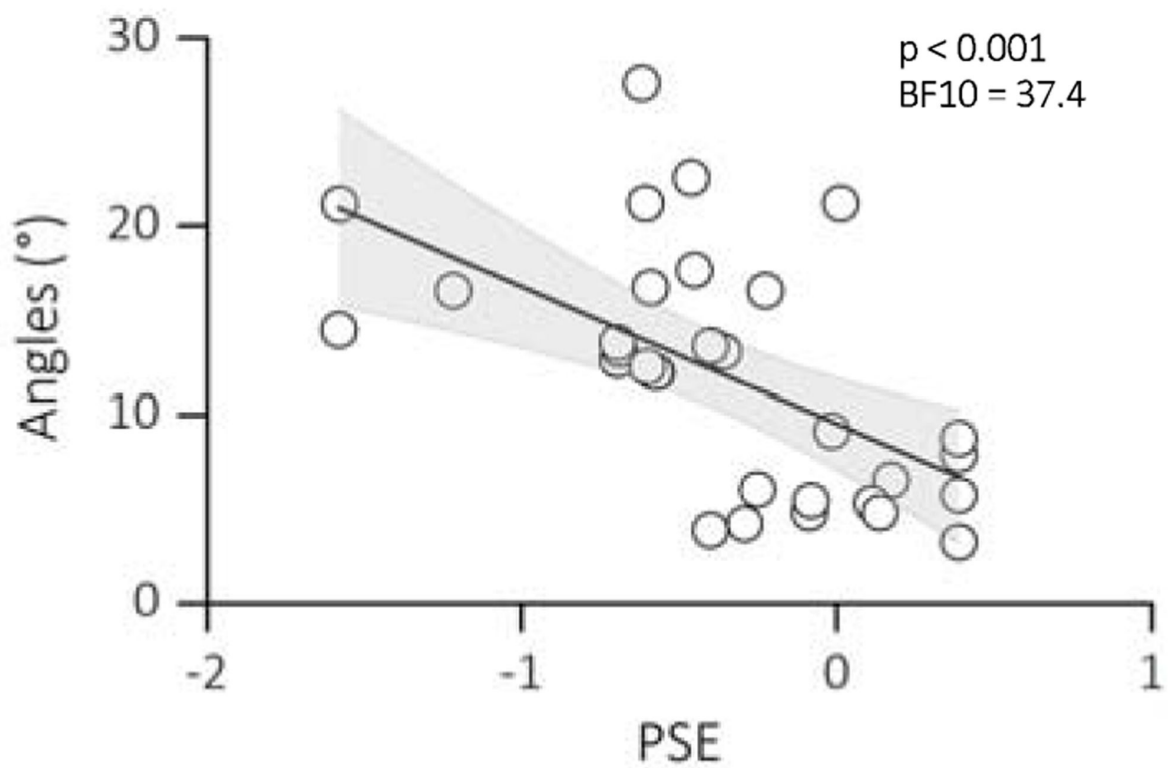


Figure 6.

Linear regression between PSEs of the trans-saccadic localization task and the angles of deviation from the double step saccade task. Text insets report p-values and associated Bayes Factors of Pearson's Rho. Thick black line shows the linear fit through the data. The shaded area represents the standard error for each observation mean as predicted by the regression line.

Discussion

In the current study we examined oculomotor efference copy signals in a sample of healthy adults with variable autistic traits using both, a motor and visual updating task. We show that a disrupted efference copy mechanism can lead to inaccurate predictions about impending eye movements, forcing the visuomotor system to rely more heavily on actual sensory input rather than the anticipated sensory consequences of self-generated eye movements.

In our first experiment, the double step saccade task, we found an impairment in the capacity of participants exhibiting high levels of autistic traits to program double-step - saccades which require efference copy information about the size of the previous saccade. Conversely, participants with relatively low autistic characteristic perform accurately in the double-step task. In the high-autism-like subsample, the final eye position after the second saccade markedly deviates from the position of the second target, and the magnitude of this error far surpasses that observed in the low-autism-like group. This specific impairment in the accuracy of second saccades lends support to the hypothesis that heightened autistic traits correlates with a deficit in constructing a spatial representation of the second target's location based on forward models of efference copy. Notably, the mislocalization of the second saccade target cannot be explained by differences in the dynamics of the saccade to the first target: The observation that the accuracy of the initial saccade in the double step task remains consistent regardless of the degree of autism-like trait severity implies that individuals with more pronounced traits do not exhibit inherent challenges in the encoding and processing of spatial information for intended saccades. Instead, when preparing for a saccade that must compensate for a preceding eye movement that altered the retinotopic representation of the target, those with more severe traits encounter difficulties. Likewise, the bias in localizing the target cannot be attributed to working memory deficits within our high-autism-like-symptom subsample; it is improbable that the mislocalization of the second target results solely from documented working memory deficits in ASD.³⁹

Our findings suggest that a general memory deficit is unlikely to fully explain the observed bias in high-AQ participants' second saccades. As highlighted in **Figure 3A**, the bias was specific to the horizontal dimension, weakening the argument for a global memory issue affecting both vertical and horizontal encoding of target location. However, it's important to acknowledge that even under non-darkness conditions, participants might rely on a combination of internal updating based on the initial target location and visual cues from the environment, such as screen borders. This potential use of visual references could contribute to the observed bias in the high-AQ group. If high-AQ participants differed in their reliance on visual cues compared to the low-AQ group, it could explain the specific pattern of altered remapping observed in the horizontal dimension. This possibility aligns with our argument for an abnormal remapping process underlying the results. While altered efference copy signals remain a strong candidate, the potential influence of visual cues on remapping in this population warrants further investigation. Future studies could incorporate a darkness condition to isolate the effects of internal updating on the first saccade, and systematically manipulate the availability of visual cues throughout the task. This would allow for a more nuanced understanding of how internal updating and visual reference use interact in the double-step paradigm, particularly for individuals with varying AQ scores.

Individuals with ASD show remarkable similarities to the symptoms of schizophrenia patients^{40–43}. Both ASD and schizophrenia patients are considered to have impairments at both extremes, involving either an excessive or inadequate reliance on sensory signals, resulting in inflexible behavior and issues in recurrent neural network representation.^{44,45} Using a double-step saccade task, previous research has reported multiple lines of evidence demonstrating

disrupted efference copy signals in schizophrenia. Importantly, similar to our findings, this disruption has implications for perceptual and motor precision, highlighting the interconnectedness of corollary discharge deficits and motor inflexibility^{46,47}.

A dramatic consequence of altered efference copy signaling, is the inability to account for own eye movements when judging the location of a target presented intra-saccadically. The trans-saccadic localization task yielded results that provide further insights into the role of efference copy in individuals with varying levels of autistic traits. In this task, all participants, regardless of their autistic trait levels, successfully executed accurate saccades to peripheral targets. However, the key factor emerged in the subsequent perceptual judgments made by these individuals. Individuals with high autistic traits exhibited a significant mislocalization of a remembered stimulus position across saccades. In essence, they perceived the visual stimulus as having shifted in space more than it actually did during the eye movement. This indicates that their brain is relying more on the actual sensory feedback than the efference copy prediction. The consequence is an inaccurate perception of the stimulus's location in space.

The results of both experiments shed light on the relationship between efference copy failure and Bayesian theories of autism^{9,10,48}. In the double step task, individuals with higher autistic traits exhibited difficulties in utilizing efference copy signals to construct a spatial representation of the second target location. This impairment suggests that in high autistic characteristics pre-saccadic predictions will never be completely confirmed by the post-saccadic feedback. This constant lack of a successful match between predicted sensory consequences of self-generated eye movements and post-saccadic evidence can lead over time to the build-up of weak priors with high variance.

Similarly, in the trans-saccadic localization task, high autistic traits were associated with failures in spatial updating across saccade execution. These findings align with the Bayesian framework, where the brain continuously updates its internal model by combining sensory information with prior knowledge. In the context of autism, a failure in efference copy can further contribute to the unique sensory integration processes observed in individuals on the autism spectrum. Our results are supported by recent evidence on the clinical population. Using a blanking task in a group of children with and without Autism, Yao et al.⁴⁹ suggested that the ability to flexibly adjust the precision of priors may serve a protective function in individuals with ASD, potentially influencing the severity of symptom presentation. Moreover, Park et al.⁵⁰ highlighted a unique relationship between smooth pursuit quality and motion prediction in ASD: unlike typically developing children, those with ASD did not exhibit a central-tendency bias over time, suggesting deficits in integrating prior knowledge about environmental statistics.

We have recently suggested a disruption in causal inference in the high autistic characteristics, as these individuals tend to attribute post-saccadic errors to external factors—a dynamic and unstable world—rather than accurately linking them to their own motor actions⁴. Consequently, a bias in causal inference emerges, stemming from the efference copy disruption, where the brain struggles to correctly attribute the cause of motor errors, leading to a heightened reliance on sensory feedback instead of motor predictions. As previously postulated this skewed causal inference also interconnect with motor inflexibility. Inflexibility in the motor domain might also have consequences for visual stability. Here, participants with high autistic traits faced challenges in executing saccades with precision, particularly when compensating for previous eye movements that had altered target representations. In the trans-saccadic localization task, we observed that these individuals leaned more heavily on post-saccadic sensory feedback than efference copy predictions. This not only led to an inaccurate perception of stimuli location but also underscored their limited ability to update and adapt motor actions based on sensory information.

It is essential to note that our participant pool lacked pre-existing diagnoses before engaging in the experiments and we must address limitations associated with the AQ questionnaire. The AQ questionnaire demonstrates adequate test-retest reliability⁵¹, normal distribution of sum scores in the general population⁵², and cross-cultural equivalence has been established in Dutch and Japanese samples^{53–55}. The AQ effectively categorizes individuals into low, average, and high degrees of autistic traits, demonstrating sensitivity for both group and individual assessments⁵⁶.

However, evolving research underscores many aspects that are not fully captured by the self-administered questionnaire: for example, gender differences in ASD trait manifestation⁵⁷. Autistic females may exhibit more socially typical interests, often overlooked by professionals⁵⁸. Camouflaging behaviors, employed by autistic women to blend in, pose challenges for accurate diagnosis⁵⁹. Late diagnoses are attributed to a lack of awareness, gendered traits, and outdated assessment tools⁶⁰. Moving forward, complementing AQ evaluations in the general population with other questionnaires, such as those assessing camouflaging abilities⁶¹, or motor skills in everyday situation (MOSES-test⁶²) becomes crucial for a comprehensive understanding of autistic traits.

In conclusion, our study underscores the critical role of efference copy in maintaining perceptual stability during eye movements and highlights its potential relevance to the sensory processing differences observed in individuals with heightened autistic traits. These findings contribute to our understanding of the complex interplay between prior knowledge, sensory integration, and motor control in the context of ASD, opening avenues for future research and potential interventions aimed at enhancing perceptual stability in this population.

Methods

Participants

Forty-two participants (31 female, mean age 23, SD = 4.19) took part to Experiment 1 (motor updating). Thirty-one of them (21 females, mean age 22.81, SD = 3.11) participated also in Experiment 2 (visual updating). Subjects were either German native speakers or English speakers with no neurological or psychiatric diseases. Participants either reported to have normal vision or they wore lenses during their acquisition. All participants were recruited through the Heinrich-Heine University Düsseldorf and received either course credit or payment of 10 euros per hour. The experimental procedure was approved by the local ethics committee of the Faculty of Mathematics and Natural Sciences of Heinrich-Heine-University, Düsseldorf (ethics approval number: PO01_2022_01).

Experimental materials and procedures

Stimuli were displayed on a 27-inch Acer XB272 LCD monitor driven by a Alienware pc (Aurora R7) with a 240-Hz refresh rate (frame duration 4.16 ms) and a resolution of 1920 × 1080 pixels. The experimental program was implemented in MATLAB 2016b (Mathworks, Natick, MA, USA) using Psychtoolbox⁶³. Eye movements and pupil diameters were recorded with the EyeLink 1000 system (SR Research Ltd., Mississauga, Ontario, Canada), which sampled eye positions at a rate of 1000 Hz. The head was sustained with a chin- and forehead-rest. For all participants the left eye was recorded. Viewing was binocular. At the beginning of each session, the EyeLink was calibrated with the standard nine-point EyeLink procedure. The system detected the start and the end of a saccade when eye velocity exceeded or fell below 30°/s.

Experiment 1: motor updating

A trial started with the presentation of a black fixation point ($0.55^\circ \times 0.55^\circ$, FP) at the screen center. After 1000 ms plus a randomly chosen duration between 0–500 ms, the first saccade target ST1 ($0.55^\circ \times 0.55^\circ$, black) appeared 10° to the right of the screen center (see [Figure 2A](#)), which remained visible for 64 ms. Upon extinction of ST1, ST2 appeared 10° upwards or downwards from ST1, and remained visible for another 64 ms. The position of ST2 was randomized across trials. ST2 and the fixation point were extinguished together, and this cued participants to start the saccade sequence: from the fixation point to ST1, and from ST1 to ST2. Participants completed 3 sessions of 100 trials each.

Experiment 2: visual updating

The trial sequence is shown in [Figure 4A](#) and [3B](#). A trial started with the presentation of a fixation point (black square FP, $0.55^\circ \times 0.55^\circ$) 5° to the left of the screen center. The fixation point stayed on for 1000 ms plus a randomly chosen duration between 0–500 ms. At a random time within the fixation point presentation time, the first stimulus was presented (Bottom Dot, BD). After disappearance of the first fixation point, participants were instructed to saccade to the second fixation point (black square FP, $0.55^\circ \times 0.55^\circ$) 5° to the right of the screen center (total saccade size 10°). At saccade completion, a second stimulus was delivered (Top Dot, TD). The stimuli consisted of two red dots (1.5° diameter), each flashed for one monitor frame with at least a temporal separation of 300 ms on average (see [Figure 4C](#) for distribution of saccade offset and second stimulus onset). The first dot could appear randomly above or below gaze level at a fixed horizontal location, halfway between the two fixations ($x = 0$, $y = -5^\circ$ or $+5^\circ$ depending on the trial). The second dot was then shown orthogonal to the first one at a variable horizontal location ($x = 0^\circ \pm 2.5^\circ$). The stimulus was perceived as a single dot moving downward or upward (depending on the position of the first dot) with a near-vertical trajectory, i.e. orthogonal to the direction of saccades. Participants completed 3 sessions of 110 trials each.

Quantification and statistical analysis

Experiment 1

In the analysis of eye movement data in Experiment 1 we excluded trials if: i) participants did not perform a saccade or they blinked during the execution of the saccade; ii) participants started the saccade sequence before the extinction of the fixation point; iii) the amplitude of the first saccade was smaller than half of the required distance, i.e. $< 5^\circ$; iv) the vertical amplitude of the second saccade was smaller than 5° ; v) participants deviated more than 2.5° on the horizontal or vertical dimension when fixating; vi) the latency of the first saccade was < 100 ms. Saccade amplitudes were averaged into 10 separate bins of ~ 30 trials each for first, and second saccades that passed the selection criteria. These criteria were applied to ensure that both saccades were large enough to reveal a putative deviation of the second saccade. If, for instance, the executed first saccade is much smaller than the required distance, the efference copy should signal a smaller amplitude, thus leading to smaller influences on the direction of the second saccade. We tested our hypothesis concerning impaired use of efference copy information in high autism phenotype by analyzing the direction of the second saccade. Specifically, we determined how far the second saccade deviated from the optimal vector that would have directed the gaze onto the target. We calculated the angle between the second saccade vector and the optimal vector connecting the starting position of the second saccade and the second saccade target (see [Figure 2B](#)). These angles were computed separately for upward and downward saccades, but since they resulted in very similar values, we collapsed the data into upward angles only.

Experiment 2

We only analyzed trials where the saccade was performed between the presentation of the two dots (first dot at least 60 ms before the saccade, second dot after its completion), therefore, in retinotopic coordinates, the two dots were always displaced horizontally by about 10° . That is when the dots were presented on the screen, they appeared at a different location on the retina, horizontally separated by an angle of approximately 10° (see also [Figure 4B](#)). In other words, each dot was perceived on the retina at a slightly different position in the visual field, due to horizontal displacement. That subjects perceived the dots displaced along a nearly vertical trajectory, indicates that the retinotopic displacement is largely compensated, ensuring spatial stability. However, small biases of reported direction can indicate relative failures in this stabilization process. To estimate biases in trans-saccadic updating, we varied the location of the second dot with the method of constant stimuli and asked subjects to report in 2AFC whether the second dot was more slanted to the right or to the left compared to the first one. Data were analyzed as psychometric functions, plotting the proportion of “rightward” judgments as function of the position of the second dot relative to the first one. Distributions were fitted with cumulative Gaussian functions; the median of the curve estimated the PSE, or the position of the second dot that led to vertical stimulus displacement. A negative bias ($PSE_{<0}$) implies a bias towards seeing rightward displacement and a positive bias ($PSE_{>0}$) implies a bias towards seeing leftward displacement (as in [Fig. 3B](#), dashed gray line). The former negative bias can be interpreted as an under-compensation of the saccade vector. Moreover, a trial was discarded if: i) participants did not perform a saccade or they blinked during the execution of the saccade; ii) participants started the saccade before the extinction of the fixation point; iii) the amplitude was smaller than half of the required distance, iv) participants deviated more than 2.5° on the horizontal or vertical dimension when fixating.

Autistic Quotient (AQ)

All participants completed the self-administered Autistic Quotient questionnaire, in the German or English validated version^{33,64}. This contains 50 items, grouped in five subscales: attention switching, attention to detail, imagination, communication, and social skills. For each question, participants read a statement and selected the degree to which the statement best described them: “strongly agree,” “slightly agree,” “slightly disagree,” and “strongly disagree”. The standard scoring described in the original paper was used: 1 when the participant’s response was characteristic of ASD (slightly or strongly), 0 otherwise. Total scores ranged between 0 and 50, with higher scores indicating higher degrees of autistic traits. All participants scored below 32, the threshold above which a clinical assessment is recommended⁵¹.

The median of the scores for the first experiment was 16, with lower and upper quartiles of 13 and 19. Scores were normally distributed, as measured by the Jarque–Bera goodness-of-fit test of composite normality ($JB = 0.98$, $p = 0.5$). The median score of the subsample participants for experiment 2 was 17, with lower and upper quartiles of 14 and 20. Scores were normally distributed, as measured by the Jarque–Bera goodness-of-fit test of composite normality ($JB = 1.39$, $p = 0.31$).

As we were interested in the effect of autistic personality traits on the results, correlation analyses were complemented with standard t-tests comparing the upper and the lower quartile of the AQ distribution’s scores for the two experiments separately. Both measures were accompanied with Bayes Factors estimation. Bayes Factors⁶⁵ quantify the evidence for or against the null hypothesis as the ratio of the likelihoods for the experimental and the null hypothesis. We express it as the ratio, where negative numbers indicate that the null hypothesis is likely to be true, positive that it is more likely false. By convention, absolute Bayes factors > 3 are considered substantial evidence for either the alternate or null hypothesis, > 10 strong evidence, and > 100 decisive.

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Author contributions

All authors contributed to the study concept and to the design. Stimuli were designed by A.P. and A.P. performed the data analysis. All authors contributed to the interpretation of results. A.P. drafted the manuscript, and E.Z. provided critical revisions. All authors approved the final version of the manuscript for submission.

Competing interests

The authors declare no competing interests.

References

1. (2013) **Diagnostic and statistical manual of mental disorders: DSM-5™** <https://doi.org/10.1176/appi.books.9780890425596>
2. Lopez B.R., Lincoln A.J., Ozonoff S., Lai Z (2005) **Examining the relationship between executive functions and restricted, repetitive symptoms of Autistic Disorder** *J Autism Dev Disord* **35**:445–460 <https://doi.org/10.1007/s10803-005-5035-x>
3. Van Eylen L., Boets B., Steyaert J., Evers K., Wagemans J., Noens I. (2011) **Cognitive flexibility in autism spectrum disorder: Explaining the inconsistencies?** *Research in Autism Spectrum Disorders* **5**:1390–1401
4. Pomè A., Tyralla S., Zimmermann E (2023) **Altered oculomotor flexibility is linked to high autistic traits** *Sci Rep* **13** <https://doi.org/10.1038/s41598-023-40044-5>
5. Johnson B.P., Rinehart N.J., White O., Millist L., Fielding J (2013) **Saccade adaptation in autism and Asperger's disorder** *Neuroscience* **243**:76–87
6. Mosconi M.W., Luna B., Kay-Stacey M., Nowinski C.V., Rubin L.H., Scudder C., Minshew N., Sweeney J.A (2013) **Saccade adaptation abnormalities implicate dysfunction of cerebellar-dependent learning mechanisms in autism spectrum disorders (ASD)** *PloS one* **8**
7. Sinha P., Kjelgaard M.M., Gandhi T.K., Tsourides K., Cardinaux A.L., Pantazis D., Diamond S.P., Held R.M (2014) **Autism as a disorder of prediction** *Proceedings of the national academy of sciences* **111**:15220–15225
8. Van Boxtel J., Lu H. (2013) **A predictive coding perspective on autism spectrum disorders** *Frontiers in Psychology* **4** <https://doi.org/10.3389/fpsyg.2013.00019>
9. Van de Cruys S., Evers K., Van der Hallen R., Van Eylen L., Boets B., de-Wit L., Wagemans J. (2014) **Precise minds in uncertain worlds: predictive coding in autism** *Psychol Rev* **121**:649–675 <https://doi.org/10.1037/a0037665>
10. Pellicano E., Burr D (2012) **When the world becomes 'too real': a Bayesian explanation of autistic perception** *Trends in cognitive sciences* **16**:504–510
11. Angeletos Chrysaitis N., Seriès P. (2023) **10 years of Bayesian theories of autism: A comprehensive review** *Neuroscience & Biobehavioral Reviews* **145** <https://doi.org/10.1016/j.neubiorev.2022.105022>
12. Cont C., Zimmermann E (2021) **The Motor Representation of Sensory Experience** *Current Biology* **31**:1029–1036 <https://doi.org/10.1016/j.cub.2020.11.032>
13. Wurtz R.H (2018) **Corollary discharge contributions to perceptual continuity across saccades** *Annual review of vision science* **4**:215–237
14. Sommer M.A., Wurtz R.H (2002) **A pathway in primate brain for internal monitoring of movements** *Science* **296**:1480–1482 <https://doi.org/10.1126/science.1069590>

15. Bellebaum, Hoffmann, K.-P., Daum I. (2005) **Post-saccadic updating of visual space in the posterior parietal cortex in humans** *Behavioural brain research* **163**:194–203
16. Bellebaum C., Daum I., Koch B., Schwarz M., Hoffmann K.-P (2005) **The role of the human thalamus in processing corollary discharge** *Brain* **128**:1139–1154
17. Bellebaum C., Hoffmann K.P., Koch B., Schwarz M., Daum I (2006) **Altered processing of corollary discharge in thalamic lesion patients** *European Journal of Neuroscience* **24**:2375–2388
18. Ostendorf F., Liebermann D., Ploner C.J (2013) **A role of the human thalamus in predicting the perceptual consequences of eye movements** *Frontiers in Systems neuroscience* **7**
19. Ostendorf F., Liebermann D., Ploner C.J (2010) **Human thalamus contributes to perceptual stability across eye movements** *Proceedings of the National Academy of Sciences* **107**:1229–1234
20. Zimmermann E., Ghio M., Pergola G., Koch B., Schwarz M., Bellebaum C (2020) **Separate and overlapping functional roles for efference copies in the human thalamus** *Neuropsychologia* **147** <https://doi.org/10.1016/j.neuropsychologia.2020.107558>
21. Burr D.C., Morrone M.C., Ross J (1994) **Selective suppression of the magnocellular visual pathway during saccadic eye movements** *Nature* **371**:511–513 <https://doi.org/10.1038/371511a0>
22. Binda P., Morrone M.C (2018) **Vision During Saccadic Eye Movements** *Annu Rev Vis Sci* **4**:193–213 <https://doi.org/10.1146/annurev-vision-091517-034317>
23. Zimmermann E., Lange J (2022) **Saccade suppression of displacements, but not of contrast, depends on context** *J Vis* **22** <https://doi.org/10.1167/jov.22.10.10>
24. Castet E., Jeanjean S., Masson G.S (2002) **Motion perception of saccade-induced retinal translation** *Proceedings of the National Academy of Sciences* **99**:15159–15163 <https://doi.org/10.1073/pnas.232377199>
25. Idrees S., Baumann M.P., Franke F., Münch T.A., Hafed Z.M (2020) **Perceptual saccadic suppression starts in the retina** *Nature Communications* **11** <https://doi.org/10.1038/s41467-020-15890-w>
26. Duhamel J.R., Colby C.L., Goldberg M.E (1992) **The updating of the representation of visual space in parietal cortex by intended eye movements** *Science* **255**:90–92 <https://doi.org/10.1126/science.1553535>
27. Westheimer G (1954) **Eye movement responses to a horizontally moving visual stimulus** *AMA archives of ophthalmology* **52**:932–941
28. Hallett P., Lightstone A (1976) **Saccadic eye movements towards stimuli triggered by prior saccades** *Vision research* **16**:99–106
29. Szinte M., Cavanagh P (2012) **Apparent Motion from Outside the Visual Field, Retinotopic Cortices May Register Extra-Retinal Positions** *PLOS ONE* **7** <https://doi.org/10.1371/journal.pone.0047386>

30. Zimmermann, Morrone, M.C., Binda P. (2018) **Perception during double-step saccades** *Scientific Reports* **8** <https://doi.org/10.1038/s41598-017-18554-w>
31. Fracasso A., Caramazza A., Melcher D (2010) **Continuous perception of motion and shape across saccadic eye movements** *J Vis* **10** <https://doi.org/10.1167/10.13.14>
32. Szinte, Cavanagh P. (2011) **Spatiotopic apparent motion reveals local variations in space constancy** *J Vis* **11** <https://doi.org/10.1167/11.2.4>
33. Ruzich E., Allison C., Smith P., Watson P., Auyeung B., Ring H., Baron-Cohen S (2015) **Measuring autistic traits in the general population: a systematic review of the Autism-Spectrum Quotient (AQ) in a nonclinical population sample of 6,900 typical adult males and females** *Molecular autism* **6**:1–12
34. Lundström S., Chang Z., Råstam M., Gillberg C., Larsson H., Anckarsäter H., Lichtenstein P (2012) **Autism Spectrum Disorders and Autisticlike Traits: Similar Etiology in the Extreme End and the Normal Variation** *Archives of General Psychiatry* **69**:46–52 <https://doi.org/10.1001/archgenpsychiatry.2011.144>
35. Robinson E.B., Koenen K.C., McCormick M.C., Munir K., Hallett V., Happé F., Plomin R., Ronald A (2011) **Evidence That Autistic Traits Show the Same Etiology in the General Population and at the Quantitative Extremes (5%, 2.5%, and 1%)** *Archives of General Psychiatry* **68**:1113–1121 <https://doi.org/10.1001/archgenpsychiatry.2011.119>
36. Bralten J., van Hulzen K.J., Martens M.B., Galesloot T.E., Arias Vasquez A., Kiemeneij L.A., Buitelaar J.K., Muntjewerff J.W., Franke B., Poelmans G. (2017) **Erratum: Autism spectrum disorders and autistic traits share genetics and biology** *Molecular Psychiatry* <https://doi.org/10.1038/mp.2017.127>
37. Sucksmith E., Roth I., Hoekstra R.A (2011) **Autistic Traits Below the Clinical Threshold: Re-examining the Broader Autism Phenotype in the 21st Century** *Neuropsychology Review* **21**:360–389 <https://doi.org/10.1007/s11065-011-9183-9>
38. Hui J., Wang Y., Zhang P., Tse P.U., Cavanagh P (2020) **Apparent Motion Is Computed in Perceptual Coordinates** *i-Perception* **11** <https://doi.org/10.1177/2041669520933309>
39. Han Y.M.Y., Chan M.-C., Chan M.M.Y., Yeung M.K., Chan A.S (2022) **Effects of working memory load on frontal connectivity in children with autism spectrum disorder: a fNIRS study** *Scientific Reports* **12** <https://doi.org/10.1038/s41598-022-05432-3>
40. Barlati S., Minelli A., Ceraso A., Nibbio G., Carvalho Silva R., Deste G., Turrina C., Vita A (2020) **Social cognition in a research domain criteria perspective: a bridge between schizophrenia and autism spectra disorders** *Frontiers in psychiatry* **11**
41. Hommer R.E., Swedo S.E. (2015) **Schizophrenia and autism—related disorders**
42. De Crescenzo F., Postorino V., Siracusano M., Riccioni A., Armando M., Curatolo P., Mazzone L. (2019) **Autistic symptoms in schizophrenia spectrum disorders: a systematic review and meta-analysis** *Frontiers in psychiatry* **10**
43. Krieger I., Grossman-Giron A., Comaneshter D., Weinstein O., Kridin K., Cohen A.D., Bitan D.T (2021) **The co-occurrence of autistic spectrum disorder and schizophrenia: A nationwide population-based study** *Journal of Psychiatric Research* **138**:280–283

44. Idei H., Murata S., Chen Y., Yamashita Y., Tani J., Ogata T (2017) **Reduced behavioral flexibility by aberrant sensory precision in autism spectrum disorder: A neurorobotics experiment** *IEEE* :271–276
45. Philippsen A., Nagai Y (2018) **Understanding the cognitive mechanisms underlying autistic behavior: a recurrent neural network study** *IEEE* :84–90
46. Thakkar K.N., Schall J.D., Logan G.D., Park S (2015) **Response inhibition and response monitoring in a saccadic double-step task in schizophrenia** *Brain Cogn* **95**:90–98 <https://doi.org/10.1016/j.bandc.2015.01.007>
47. Thakkar K.N., Schall J.D., Heckers S., Park S (2015) **Disrupted Saccadic Corollary Discharge in Schizophrenia** *The Journal of Neuroscience* **35**:9935–9945 <https://doi.org/10.1523/jneurosci.0473-15.2015>
48. Lawson R.P., Rees G., Friston K.J (2014) **An aberrant precision account of autism** *Frontiers in human neuroscience* **8**
49. Yao B., Rolfs M., McLaughlin C., Isenstein E.L., Guillory S.B., Grosman H., Kashy D.A., Foss-Feig J.H., Thakkar K.N (2021) **Oculomotor corollary discharge signaling is related to repetitive behavior in children with autism spectrum disorder** *J Vis* **21** <https://doi.org/10.1167/jov.21.8.9>
50. Park W.J., Schauder K.B., Kwon O.S., Bennetto L., Tadin D (2021) **Atypical visual motion prediction abilities in autism spectrum disorder** *Clin Psychol Sci* **9**:944–960 <https://doi.org/10.1177/2167702621991803>
51. Baron-Cohen S., Wheelwright S., Skinner R., Martin J., Clubley E (2001) **The autism-spectrum quotient (AQ): evidence from Asperger syndrome/high-functioning autism, males and females, scientists and mathematicians** *J Autism Dev Disord* **31**:5–17 <https://doi.org/10.1023/a:1005653411471>
52. Hurst R.M., Mitchell J.T., Kimbrel N.A., Kwapił T.K., Nelson-Gray R.O (2007) **Examination of the reliability and factor structure of the Autism Spectrum Quotient (AQ) in a non-clinical sample** *Personality and Individual Differences* **43**:1938–1949 <https://doi.org/10.1016/j.paid.2007.06.012>
53. Hoekstra R.A., Bartels M., Cath D.C., Boomsma D.I (2008) **Factor structure, reliability and criterion validity of the Autism-Spectrum Quotient (AQ): a study in Dutch population and patient groups** *J Autism Dev Disord* **38**:1555–1566 <https://doi.org/10.1007/s10803-008-0538-x>
54. Kurita H., Koyama T., Osada H (2005) **Autism-Spectrum Quotient-Japanese version and its short forms for screening normally intelligent persons with pervasive developmental disorders** *Psychiatry Clin Neurosci* **59**:490–496 <https://doi.org/10.1111/j.1440-1819.2005.01403.x>
55. Wakabayashi A., Baron-Cohen S., Wheelwright S., Tojo Y (2006) **The Autism-Spectrum Quotient (AQ) in Japan: a cross-cultural comparison** *Journal of autism and developmental disorders* **36**:263–270
56. Tennant A., Conaghan P.G (2007) **The Rasch measurement model in rheumatology: what is it and why use it? When should it be applied, and what should one look for in a Rasch paper?** *Arthritis Care & Research* **57**:1358–1362

57. Lai M.C., Lombardo M.V., Pasco G., Ruigrok A.N., Wheelwright S.J., Sadek S.A., Chakrabarti B., Baron-Cohen S (2011) **A behavioral comparison of male and female adults with high functioning autism spectrum conditions** *PLoS One* **6** <https://doi.org/10.1371/journal.pone.0020835>
58. Simcoe S.M., Gilmour J., Garnett M.S., Attwood T., Donovan C., Kelly A.B (2023) **Are there gender-based variations in the presentation of Autism amongst female and male children?** *Journal of Autism and Developmental Disorders* **53**:3627–3635 <https://doi.org/10.1007/s10803-022-05552-9>
59. Cook J., Crane L., Hull L., Bourne L., Mandy W (2022) **Self-reported camouflaging behaviours used by autistic adults during everyday social interactions** *Autism* **26**:406–421 <https://doi.org/10.1177/13623613211026754>
60. Zener D (2019) **Journey to diagnosis for women with autism** *Advances in Autism* **5**:2–13 <https://doi.org/10.1108/AIA-10-2018-0041>
61. Hull L., Mandy W., Lai M.-C., Baron-Cohen S., Allison C., Smith P., Petrides K.V (2019) **Development and Validation of the Camouflaging Autistic Traits Questionnaire (CAT-Q)** *Journal of Autism and Developmental Disorders* **49**:819–833 <https://doi.org/10.1007/s10803-018-3792-6>
62. Hillus J., Moseley R., Roepke S., Mohr B (2019) **Action Semantic Deficits and Impaired Motor Skills in Autistic Adults Without Intellectual Impairment** *Frontiers in Human Neuroscience* **13** <https://doi.org/10.3389/fnhum.2019.00256>
63. Kleiner M., Brainard D.H., Pelli D., Ingling A., Murray R., Broussard C (2007) **What's new in Psychtoolbox-3** *Perception* **36**:1–16 <https://doi.org/10.1068/v070821>
64. Freitag C.M., Retz-Junginger P., Retz W., Seitz C., Palmason H., Meyer J., Rösler M., von Gontard A. (2007) **Evaluation der deutschen version des Autismus-Spektrum-Quotienten (AQ)-die Kurzversion AQ-k** *Zeitschrift für klinische Psychologie und Psychotherapie* **36**:280–289
65. Rouder J.N., Speckman P.L., Sun D., Morey R.D., Iverson G (2009) **Bayesian t tests for accepting and rejecting the null hypothesis** *Psychonomic Bulletin & Review* **16**:225–237 <https://doi.org/10.3758/PBR.16.2.225>

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Reviewer #1 (Public Review):

Summary:

This study examines a hypothesized link between autism symptomatology and efference copy mechanisms. This is an important question for a number of reasons. Efference copy is both a critical brain mechanism that is key to rapid sensorimotor behaviors, and one that has

important implications for autism given recent empirical and theoretical work implicating atypical prediction mechanisms and atypical reliance on priors in ASD. The authors test this relationship in two different experiments, both of which show larger errors/biases in spatial updating for those with heightened autistic traits (as measured by AQ in neurotypical (NT) individuals).

Strengths:

The empirical results are convincing - effects are strong, sample sizes are sufficient, and the authors also rule out alternative explanations (ruling out differences in motor behavior or perceptual processing per se).

Weaknesses:

My main residual concern is that the paper should be more transparent about both (1) that this study does not include individuals with autism, and (2) acknowledging the limitations of the AQ.

On the first point, and I don't think this is intentional, there are several instances where the line between heightened autistic traits in the NT population and ASD is blurred or absent. For example, in the second sentence of the abstract, the authors state "Here, we examine the idea that sensory overload in ASD may be linked to issues with efference copy mechanisms". I would say this is not correct because the authors did not test individuals with ASD. I don't see a problem with using ASD to motivate and discuss this work, but it should be clear in key places that this was done using AQ in NT individuals.

For the second issue, the AQ measure itself has some problems. For example, reference 38 in the paper (a key AQ paper) also shows that the AQ is skewed more male than modern estimates of ASD, suggesting that the AQ may not fully capture the full spectrum of ASD symptomatology.

Of course, this does not mean that the AQ is not a useful measure (the present data clearly show that it captures something important about spatial updating during eye movements), but it should not be confused with ASD, and its limitations need to be acknowledged. My recommendation would be to do this in the title as well - e.g. note impaired visuomotor updating in individuals with "heightened autistic traits".

Suggestions for improvement:

- Figure 5 is really interesting. I think it should be highlighted a bit more, perhaps even with a model that uses the results of both tasks to predict AQ scores.
- Some discussion of the memory demands of the tasks will be helpful. The authors argue that memory is not a factor, but some support for this is needed.
- With 3 sessions for each experiment, the authors also have data to look at learning. Did people with high AQ get better over time, or did the observed errors/biases persist throughout the experiment?

<https://doi.org/10.7554/eLife.94946.2.sa2>

Reviewer #2 (Public Review):

Summary:

The idea that various clinical conditions may be associated, at least partially, with a disrupted corollary discharge mechanism has been present for long. In this paper, the authors draw a link between sensory overload, a characteristic of autism spectrum disorder, and a disturbance in the corollary discharge mechanism. The authors substantiate their hypothesis with strong evidence from both the motor and perceptual domains. As a result, they broaden

the clinical relevance of the corollary discharge mechanism to encompass autism spectrum disorder.

Public comments:

The authors write:

"Imagine a scenario in which you're watching a video of a fast-moving car on a bumpy road. As the car hits a pothole, your eyes naturally make quick, involuntary saccades to keep the car in your visual field. Without a functional efference copy system, your brain would have difficulty accurately determining the current position of your eye in space, which in turn affects its ability to anticipate where the car should appear after each eye movement."

I appreciate the use of examples to clarify the concept of efference copy. However, I believe this example is more related to a gain-field mechanism, informing the system about the position of the eye with respect to the head, rather than an example of efference copy per-se.

Without an efference copy mechanism, the brain would have trouble to accurately determine where the eyes will be in space after an eye movement, and it will have trouble predicting the sensory consequences of the eye movement. But it can be argued that the gain-field mechanism would be sufficient to inform the brain about the current position of the eyes with respect the head.

The authors write:

"In the double-step paradigm, two consecutive saccades are made to briefly displayed targets 21,22. The first saccade occurs without visual references, relying on internal updating to determine the eye's position."

Maybe I am missed something, but in the double-step paradigm the first saccade can occur without the help of visual references if no visual feedback is present, that is, when saccades are performed in total darkness. Was this the case for this experiment? I could not find details about room conditions in the methods. Please provide further details.

In case saccades were not performed in total darkness, then the first saccade can be based on the remembered location of the first target presented, which can be derived from the retinotopic trace of the first stimuli, as well as contribution from the surroundings, that is: the remembered relative location of the first target with respect to the screen border along the horizontal meridian (i.e. allocentric cues)

A similar logic could be applied to the second saccade. If the second saccade were based only on the retinotopic trace, without updating, then it would go up and 45 deg to the right, based on the example shown in Figure 1. With appropriate updating, the second saccade would go straight up. However, if saccades were not performed in total darkness, then the location of the second target could also be derived from its relationship with the surroundings (for example, the remembered distance from screen borders, i.e. allocentric cues).

If saccades were not performed in total darkness, the results shown in Figures 2 and 3 could then be related to: i) differences in motor updating between AQ score groups; ii) differences in the use of allocentric cues between AQ score groups; iii) a combination of i) and ii). I believe this is a point worth mentioning in the discussion."

The authors write:

"According to theories of saccadic suppression, an efference copy is necessary to predict the occurrence of a saccade."

I would also refer to alternative accounts, where saccadic suppression appears to arise as early as the retina, due to the interaction between the visual shift introduced by the eye movement, and the retinal signal associated with the probe used to measure saccadic

suppression. This could potentially account for the scaling of saccadic suppression magnitude with saccade amplitude.

Idrees, S., Baumann, M.P., Franke, F., Münch, T.A. and Hafed, Z.M., 2020. Perceptual saccadic suppression starts in the retina. *Nature communications*, 11(1), p.1977.

<https://doi.org/10.7554/eLife.94946.2.sa1>

Reviewer #3 (Public Review):

Summary:

This work examined efference copy related to eye movements in healthy adults who have high autistic traits. Efference copies allow the brain to make predictions about sensory outcomes of self-generated actions, and thus serve important roles in motor planning and maintaining visual stability. Consequently, disrupted efference copies have been posited as a potential mechanism underlying motor and sensory symptoms in psychopathology such as Autism Spectrum Disorder (ASD), but so far very few studies have directly investigated this theory. Therefore, this study makes an important contribution as an attempt to fill in this knowledge gap. The authors conducted two eye-tracking experiments examining the accuracy of motor planning and visual perception following a saccade, and found that participants with high autistic traits exhibited worse task performance (i.e., less accurate second saccade and biased perception of object displacement), consistent with their hypothesis of less impact of efference copies on motor and visual updating. Moreover, the motor and visual biases are positively correlated, indicative of a common underlying mechanism. These findings are promising and can have important implications for clinical intervention, if they can be replicated in a clinical sample.

Strengths:

The authors utilized well-established and rigorously designed experiments and sound analytic methods. This enables easy translations between similar work in non-human primates and humans and readily points to potential candidates for underlying neural circuits that could be further examined in follow-up studies (e.g., superior colliculus, frontal eye fields, mediodorsal thalamus). The finding of no association between initial saccade accuracy and level of autistic trait in both experiments also serves as an important control analysis and increases one's confidence in the conclusion that the observed differences in task performance were indeed due to disrupted efference copies, not confounding factors such as basic visual/motor deficits or issues with working memory. The strong correlation between the observed motor and visual biases further strengthens the claim that the findings from both experiments may be explained by the same underlying mechanism - disrupted efference copies. Lastly, the authors also presented a thoughtful and detailed mechanistic theory of how efference copy impairment may lead to ASD symptomatology, which can serve as a nice framework for more research into the role of efference copies in ASD.

Weaknesses:

Although the paper has a lot of strengths, the main weakness of the paper is that a direct link with sensory/motor symptoms cannot be established. As the authors have discussed, the most likely symptoms resulting from disrupted efference copies would be sensory overload and motor inflexibility. The measure used to quantify the level of autistic traits, Autistic Quotient (AQ), does not capture any sensory or motor characteristics of the Autism spectrum. Therefore, it is unknown whether those scored high on AQ in this study experienced high, or even any, sensory or motor difficulties. In other words, more evidence is needed to

demonstrate a direct link between disrupted efference copies and sensory/motor symptoms in ASD.

<https://doi.org/10.7554/eLife.94946.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

This important study tests the hypothesis that a high autism quotient in neurotypical adults is strongly associated with suboptimal motor planning and visual updating after eye movements, which in turn, is related to a disrupted efference copy mechanism. The implication is that such abnormal behavior would be exaggerated in those with ASD and may contribute to sensory overload - a key symptom in this condition. The evidence presented is convincing, with significant effects in both visual and motor domains, adequate sample sizes, and consideration of alternatives. However, the study would be strengthened with minor but necessary corrections to methods and statistics, as well as a moderation of claims regarding direct application to ASD in the absence of testing such patients.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This study examines a hypothesized link between autism symptomatology and efference copy mechanisms. This is an important question for several reasons. Efference copy is both a critical brain mechanism that is key to rapid sensorimotor behaviors, and one that has important implications for autism given recent empirical and theoretical work implicating atypical prediction mechanisms and atypical reliance on priors in ASD.

The authors test this relationship in two different experiments, both of which show larger errors/biases in spatial updating for those with heightened autistic traits (as measured by AQ in neurotypical (NT) individuals).

Strengths:

The empirical results are convincing - effects are strong, sample sizes are sufficient, and the authors also rule out alternative explanations (ruling out differences in motor behavior or perceptual processing per se).

Weaknesses:

My main concern is that the paper should be more transparent about both (1) that this study does not include individuals with autism, and (2) acknowledging the limitations of the AQ.

On the first point, and I don't think this is intentional, there are several instances where the line between heightened autistic traits in the NT population and ASD is blurred or absent. For example, in the second sentence of the abstract, the authors state "Here, we examine the idea that sensory overload in ASD may be linked to issues with efference copy mechanisms". I would say this is not correct because the authors did not test individuals with ASD. I don't see a problem with using ASD to motivate and discuss this work, but it should be clear in key places that this was done using AQ in NT individuals.

For the second issue, the AQ measure itself has some problems. For example, reference 38 in the paper (a key paper on AQ) also shows that those with high AQ skew more male than modern estimates of ASD, suggesting that the AQ may not fully capture the full spectrum of ASD symptomatology. Of course, this does not mean that the AQ is not a useful measure (the present data clearly show that it captures something important about spatial updating during eye movements), but it should not be confused with ASD, and its limitations need to be acknowledged. My recommendation would be to do this in the title as well - e.g. note impaired visuomotor updating in individuals with "heightened autistic traits".

We thank the reviewer for the kind words. We now specify more carefully that our sample of participants consists of neurotypical adults scored for autistic traits and none of them was diagnosed with autism before participating in our experiment. Regarding the Autistic Quotient Questionnaire (AQ) on page 5 of the Introduction we now write:

"The autistic traits of the whole population form a continuum, with ASD diagnosis usually situated on the high end 31-33. Moreover, autistic traits share a genetic and biological etiology with ASD 34. Thus, quantifying autistic-trait-related differences in healthy people can provide unique perspectives as well as a useful surrogate for understanding the symptoms of ASD 31,35."

In the Discussion (page 9) we now write:

"It is essential to note that our participant pool lacked pre-existing diagnoses before engaging in the experiments and we must address limitations associated with the AQ questionnaire. The AQ questionnaire demonstrates adequate test-retest reliability 36, normal distribution of sum scores in the general population 50, and cross-cultural equivalence has been established in Dutch and Japanese samples 51-53. The AQ effectively categorizes individuals into low, average, and high degrees of autistic traits, demonstrating sensitivity for both group and individual assessments 54.

However, evolving research underscores many aspects that are not fully captured by the self-administered questionnaire: for example, gender differences in ASD trait manifestation 55. Autistic females may exhibit more socially typical interests, often overlooked by professionals 56. Camouflaging behaviors, employed by autistic women to blend in, pose challenges for accurate diagnosis 57. Late diagnoses are attributed to a lack of awareness, gendered traits, and outdated assessment tools 58. Moving forward, complementing AQ evaluations in the general population with other questionnaires, such as those assessing camouflaging abilities 59, or motor skills in everyday situation (MOSES-test 60) becomes crucial for a comprehensive understanding of autistic traits."

Suggestions for improvement:

- Figure 5 is really interesting. I think it should be highlighted a bit more, perhaps even with a model that uses the results of both tasks to predict AQ scores.

We thank the reviewer for the suggestion. However, the sample size is relatively small for building a robust and generalizable model to predict AQ scores. Statistical models built on small datasets can be prone to overfitting, meaning that they might not accurately predict the AQ for new individuals.

- Some discussion of the memory demands of the tasks will be helpful. The authors argue that memory is not a factor, but some support for this is needed.

The reviewer raises an important point regarding the potential for memory demands to influence our results. We have now also investigated the accuracy of the second saccade separately for the x and y dimension. As also shown in figure 3 panel A, a motor bias was observed only in one dimension (x), weakening the argument of memory which would imply a bias in both directions (participants remembering the position of the target relative to both screen borders for example). We performed a t-test between our subsample of participants and indeed we found a difference in saccade accuracy for the x dimension ($p = 0.03$) but not in the y dimension ($p = 0.88$).

We now add these analyses in Discussion on page 8.

- With 3 sessions for each experiment, the authors also have data to look at learning. Did people with high AQ get better over time, or did the observed errors/biases persist throughout the experiment?

We thank the reviewer for pointing this out. On page 7 (Results) we now write:

” Understanding how these biases might change over time could provide further insights into this mechanism. Specifically, we investigated whether participants exhibited any learning effects throughout the experiments. For data of Experiment 1 – motor updating – we divided our data into 10 separate bins of 30 trials each. We conducted a repeated measure ANOVA with the within-subject factor “number of sessions” (two main sessions of 5 bins each, ~150 trials) and the between-subject factor “group” (lower vs upper quartile of the AQ distribution). We found no main effect of “number of sessions” ($F(1,7) = 0.25$, $p = 0.66$), a main effect of “group” ($F(1,7) = 2.52$, $p = 0.015$), and no interaction between the two subsample of participants and the sessions tested ($F(1,7) = 0.51$, $p = 0.49$). Data of Experiment 2 – visual updating – were separated into 3 sessions. For each session we extracted the PSE and we conducted a repeated measure ANOVA with within subject factor “sessions” and between subject factor “groups” (lower vs upper quartile of the AQ distribution). Also here we found no main effect of sessions ($F(1,13) = 0.86$, $p = 0.39$), a main effect of group ($F(1,14) = 11.85$, $p = 0.004$), and no interaction between the two subsample of participants and the sessions tested ($F(1,13) = 0.20$, $p = 0.73$). In conclusion, the current study found no evidence of learning effects across the experimental sessions. However, a significant main effect of group was observed in both Experiment 1 (motor updating) and Experiment 2 (visual updating). Participants in the group with higher autistic traits performed systematically differently on the task, regardless of the number of sessions completed compared to those in the group with lower autistic traits.”

Reviewer #2 (Public Review):

Summary:

The idea that various clinical conditions may be associated, at least partially, with a disrupted corollary discharge mechanism has been present for a long time.

In this paper, the authors draw a link between sensory overload, a characteristic of autism spectrum disorder, and a disturbance in the corollary discharge mechanism. The authors substantiate their hypothesis with strong evidence from both the motor and perceptual domains. As a result, they broaden the clinical relevance of the corollary discharge mechanism to encompass autism spectrum disorder.

The authors write:

"Imagine a scenario in which you're watching a video of a fast-moving car on a bumpy road. As the car hits a pothole, your eyes naturally make quick, involuntary saccades to keep the car in your visual field. Without a functional efference copy system, your brain

would have difficulty accurately determining the current position of your eye in space, which in turn affects its ability to anticipate where the car should appear after each eye movement."

I appreciate the use of examples to clarify the concept of efference copy. However, I believe this example is more related to a gain-field mechanism, informing the system about the position of the eye with respect to the head, rather than an example of efference copy per se.

Without an efference copy mechanism, the brain would have trouble accurately determining where the eyes will be in space after an eye movement, and it will have trouble predicting the sensory consequences of the eye movement. However it can be argued that the gain-field mechanism would be sufficient to inform the brain about the current position of the eyes with respect to the head.

We now used a different example. And on page 3 of Introduction, we now write:

"During a tennis game, rapid oculomotor saccades are employed to track the high-velocity ball across the visual display. In the absence of a functional efference copy mechanism, the brain would encounter difficulty in anticipating the precise retinal location of the ball following each saccade. This could result in a transient period of visual disruption as the visual system adjusts to the new eye position. The efference copy, by predicting the forthcoming sensory consequences of the saccade, would bridge this gap and facilitate the maintenance of a continuous and accurate representation of the ball's trajectory."

The authors write:

"In the double-step paradigm, two consecutive saccades are made to briefly displayed targets 21, 22. The first saccade occurs without visual references, relying on internal updating to determine the eye's position."

Maybe I have missed something, but in the double-step paradigm the first saccade can occur without the help of visual references if no visual feedback is present, that is, when saccades are performed in total darkness. Was this the case for this experiment? I could not find details about room conditions in the methods. Please provide further details.

In case saccades were not performed in total darkness, then the first saccade can be based on the remembered location of the first target presented, which can be derived from the retinotopic trace of the first stimuli, as well as the contribution from the surroundings, that is: the remembered relative location of the first target with respect to the screen border along the horizontal meridian (i.e. allocentric cues).

A similar logic could be applied to the second saccade. If the second saccade were based only on the retinotopic trace, without updating, then it would go up and 45 deg to the right, based on the example shown in Figure 1. With appropriate updating, the second saccade would go straight up. However, if saccades were not performed in total darkness, then the location of the second target could also be derived from its relationship with the surroundings (for example, the remembered distance from screen borders, i.e. allocentric cues).

If saccades were not performed in total darkness, the results shown in Figures 2 and 3 could then be related to i) differences in motor updating between AQ score groups; ii) differences in the use of allocentric cues between AQ score groups; iii) a combination of i) and ii). I believe this is a point worth mentioning in the discussion."

Thank you for raising the important issue of visual references in the double-step saccade task. Participants performed saccades in a dimly lit room where visual references, i.e. the

screen borders, were barely visible. At the time we collected the data a laboratory that allowed performing experiments in complete darkness was not at our disposal. We acknowledge the possibility that participants could have memorized the target locations relative to the screen borders. The bias of high AQ participants could then be attributed to differences in either encoding, memorization or decoding of the target location relative to the screen borders. However, the potentially abnormal use of visual references must reflect an altered remapping process since we did not find differences in saccade landing in the vertical dimension. A t-test between our group of participants revealed a difference in saccade accuracy for the x dimension ($p = 0.03$) but not in the y dimension ($p = 0.88$). We thus agree that in addition to an altered efference copy signal in high AQ participants, altered use of visual references might also affect their saccadic remapping.

In Discussion we now write: “Our findings suggest that a general memory deficit is unlikely to fully explain the observed bias in high-AQ participants' second saccades. As highlighted in Figure 3A, the bias was specific to the horizontal dimension, weakening the argument for a global memory issue affecting both vertical and horizontal encoding of target location. However, it's important to acknowledge that even under non-darkness conditions, participants might rely on a combination of internal updating based on the initial target location and visual cues from the environment, such as screen borders. This potential use of visual references could contribute to the observed bias in the high-AQ group. If high-AQ participants differed in their reliance on visual cues compared to the low-AQ group, it could explain the specific pattern of altered remapping observed in the horizontal dimension. This possibility aligns with our argument for an abnormal remapping process underlying the results. While altered efference copy signals remain a strong candidate, the potential influence of visual cues on remapping in this population warrants further investigation. Future studies could incorporate a darkness condition to isolate the effects of internal updating on the first saccade, and systematically manipulate the availability of visual cues throughout the task. This would allow for a more nuanced understanding of how internal updating and visual reference use interact in the double-step paradigm, particularly for individuals with varying AQ scores “.

The authors write:

According to theories of saccadic suppression, an efference copy is necessary to predict the occurrence of a saccade."

I would also refer to alternative accounts, where saccadic suppression appears to arise as early as the retina, due to the interaction between the visual shift introduced by the eye movement, and the retinal signal associated with the probe used to measure saccadic suppression. This could potentially account for the scaling of saccadic suppression magnitude with saccade amplitude.

Idrees, S., Baumann, M.P., Franke, F., Münch, T.A. and Hafed, Z.M., 2020. Perceptual saccadic suppression starts in the retina. Nature communications, 11(1), p.1977.

We thank the reviewer. Now on page 4 of Introduction we write:

“Some theories consider saccadic omission and saccadic suppression as resulting from an active mechanism. In this view an efference copy would signal the occurrence of a saccade, yielding a transient decrease in visual sensitivity²⁰⁻²². Others however have pointed out the possibility that a purely passive mechanism suffices to induce saccadic omission²³. A recent study has found evidence for saccadic suppression already in the retina. Idrees et al.²⁴ demonstrated that retinal ganglion cells in isolated retinæ of mice and pigs respond to saccade-like displacements, leading to the suppression of responses to additional flashed visual stimuli through visually triggered retinal-circuit mechanisms. Importantly, their findings suggest that perisaccadic modulations of contrast sensitivity may have a purely

visual origin, challenging the need for an efference copy in the early stages of saccadic suppression. However, the suppression they measured lasted much longer than time-courses observed in behavioral data. An efference copy signal could thus be necessary to release perception from suppression.”

Reviewer #3 (Public Review):

Summary:

This work examined efference copy related to eye movements in healthy adults who have high autistic traits. Efference copies allow the brain to make predictions about sensory outcomes of self-generated actions, and thus serve important roles in motor planning and maintaining visual stability. Consequently, disrupted efference copies have been posited as a potential mechanism underlying motor and sensory symptoms in psychopathology such as Autism Spectrum Disorder (ASD), but so far very few studies have directly investigated this theory. Therefore, this study makes an important contribution as an attempt to fill in this knowledge gap. The authors conducted two eye-tracking experiments examining the accuracy of motor planning and visual perception following a saccade and found that participants with high autistic traits exhibited worse task performance (i.e., less accurate second saccade and biased perception of object displacement), consistent with their hypothesis of less impact of efference copies on motor and visual updating. Moreover, the motor and visual biases are positively correlated, indicative of a common underlying mechanism. These findings are promising and can have important implications for clinical intervention if they can be replicated in a clinical sample.

Strengths:

The authors utilized well-established and rigorously designed experiments and sound analytic methods. This enables easy translations between similar work in non-human primates and humans and readily points to potential candidates for underlying neural circuits that could be further examined in follow-up studies (e.g., superior colliculus, frontal eye fields, mediodorsal thalamus). The finding of no association between initial saccade accuracy and level of autistic trait in both experiments also serves as an important control analysis and increases one's confidence in the conclusion that the observed differences in task performance were indeed due to disrupted efference copies, not confounding factors such as basic visual/motor deficits or issues with working memory. The strong correlation between the observed motor and visual biases further strengthens the claim that the findings from both experiments may be explained by the same underlying mechanism - disrupted efference copies. Lastly, the authors also presented a thoughtful and detailed mechanistic theory of how efference copy impairment may lead to ASD symptomatology, which can serve as a nice framework for more research into the role of efference copies in ASD.

Weaknesses:

Although the paper has a lot of strengths, the main weakness of the paper is that a direct link with ASD symptoms (i.e., sensory overload and motor inflexibility as the authors suggested) cannot be established. First of all, the participants are all healthy adults who do not meet the clinical criteria for an ASD diagnosis. Although they could be considered a part of the broader autism phenotype, the results cannot be easily generalized to the clinical population without further research. Secondly, the measure used to quantify the level of autistic traits, Autistic Quotient (AQ), does not actually capture any sensory or motor symptoms of ASD. Therefore, it is unknown whether those who scored high on AQ in this study experienced high, or even any, sensory or motor difficulties. In other words,

more evidence is needed to demonstrate a direct link between disrupted efference copies and sensory/motor symptoms in ASD.

This is a valid point, and we thank the reviewer for raising it up. Moving forward, complementing AQ evaluations in the general population with other questionnaires, such as those assessing camouflaging abilities (Hull, L., Mandy, W., Lai, MC., et al., 2019), or motor skills in everyday situation (MOSES-test, Hillus J, Moseley R, Roepke S, Mohr B. 2019) becomes crucial for a comprehensive understanding of autistic traits.”

We now address this point in Discussion page 9.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Minor comments

- The pothole example in the introduction was really hard to follow. I wonder if there is a better example.

We now used a different example. And on page 3 of Introduction, we now write:

“During a tennis game, rapid oculomotor saccades are employed to track the high-velocity ball across the visual display. In the absence of a functional efference copy mechanism, the brain would encounter difficulty in anticipating the precise retinal location of the ball following each saccade. This could result in a transient period of visual disruption as the visual system adjusts to the new eye position. The efference copy, by predicting the forthcoming sensory consequences of the saccade, would bridge this gap and facilitate the maintenance of a continuous and accurate representation of the ball's trajectory.”

- This is really minor; I would say that saccades are not the most frequent movement that humans perform. Some of the balance-related adjustments and even heartbeats are faster. Maybe just add "voluntary".

We thank the reviewer for the suggestion, now added.

- "Severe consequences" on page 4 is a bit strong. If that were true, there would be pretty severe impairments in eye movement behavior in ASD, which I don't think is the case.

We agree with the reviewer. We now eliminated the term “severe”.

- The results section would read better if each experiment had a short paragraph reiterating its overall goal and the specific approach each experiment took to achieve that goal.

Now on page 5, for the first experiment, we write:

“We investigated the influence of autistic traits on visual updating during saccadic eye movements using a classic double-step saccade task. This task relies on participants making two consecutive saccades to briefly presented targets. The accuracy of the second saccade serves as an indirect measure of how effectively the participant's brain integrated the execution of the first saccade into their internal representation of visual space. Participants were divided into quartiles based on the severity of their autistic traits, as assessed by the Autistic quotient questionnaire (cite). We hypothesized that individuals with higher autistic traits would exhibit greater difficulty in visual updating compared to those with lower autistic traits. This would be reflected in reduced accuracy of their second saccades in the

double-step task. **Figure 2C** illustrates examples from participants at the extremes of the autistic trait distribution (Autistic quotient = 3, in orange and Autistic quotient = 31, in magenta). As shown, both participants were instructed to make saccades to the locations indicated by two brief target appearances (T1 and T2), as quickly and accurately as possible, following the order of presentation. However, successful execution of the second saccade requires accurate internal compensation for the first saccade, without any visual references or feedback available during the saccade itself.”

On page 6, for experiment 2, we write:

”With a trans-saccadic localization task, we explored how autistic traits affect the integration of eye movements into visual perception. Participants were presented with stimuli before and after a single saccade, creating an illusion of apparent motion. We measured the perceived direction of this displacement, which is influenced by how well the participant's brain accounts for the saccadic eye movement. We predicted that individuals with higher autistic traits would show a stronger bias in the perceived displacement direction, suggesting a less accurate integration of the eye movement into their visual perception.”

- On page 6, the text about "vertical displacement" is confusing. The spatial displacements in this experiment were horizontal?

Yes, they were. The spatial displacement is horizontal, but the perceived trajectory (due to the saccade) is vertical. We now changed “vertical displacement” to “vertical trajectory”.

- Page 6, grammatical problems in "while we report a slightly slant of the dots trajectory".

Thank you. Now fixed.

- It would be helpful to discuss the apparent motion part of Experiment 2 in the main text. This important part is not made clear.

We now in Introduction, page 4, write:

“In this paradigm, one stimulus is shown before and another after saccade execution. Together these two stimuli produce the perception of “apparent motion”. If stimuli are placed such that the apparent motion path is orthogonal to the saccade path, then the orientation of the apparent motion path indicates how the saccade vector is integrated into vision. The apparent motion trajectory can only appear vertical if the movement of the eyes is perfectly accounted for, that is the retinotopic displacement is largely compensated, ensuring spatial stability. However, small biases of motion direction – implying under- (or over-) compensation of the eye movement – can indicate relative failures in this stabilization process. In a seminal study, Szinte and Cavanagh 27 found a slight over-compensation of the saccade vector leading to apparent motion slightly tilted against the direction of the saccade. More importantly, when efference copies are not available, i.e. localization occurring at the time of a second saccade in a double step task, a strong saccade under-compensation occurs 28.

This phenomenon cannot be explained by perisaccadic mislocalization of flashed visual stimuli 29,30, but the two phenomena may be related in that they may both depend upon efference copy information.”

- Figure 1 could be improved. For example, the text talks about the motor plan, but this is not clearly shown in the figure.

We now added the motor plan into the model. Thank you.

- Figure 2A, the scale is off (the pictures make it look like the horizontal movement was longer than the vertical).

Now fixed.

- Figure 4, it would be helpful if the task was also described in the figure.

We thank the reviewer for the comment. We now tried to modify the figure by also adding the perceptual judgment task.

- Figure 5A, the y-axis shows $p(\text{correct})$, but that is not what the y-axis shows (the legend makes the same mistake).

We apologize, it's the proportion of time participants reported the second dot to be more to the right compared to the first one. We now changed the figure and the text accordingly.

- A recent study on motion and eye movement prediction in ASD is very relevant to the work presented here.: Park et al. (2021). Atypical visual motion-prediction abilities in autism spectrum disorder. *Clinical Psychological Science*, 9(5), 944-960.

Indeed. We now refer to the cited study in Discussion, on page 9.

Reviewer #2 (Recommendations For The Authors):

Statistics and plotting.

I believe some of the reported statistics are not clear. For example, the authors write:

"Saccade landing positions of participants in the lower quartile (mean degree {plus minus} SEM: 10.17{plus minus} 0.50) did not deviate significantly from those in the upper quartile (mean degree {plus minus} SEM: 9.65 {plus minus} 0.77). This result was also confirmed by a paired sample t-test ($t(7) = 0.66$; $p = 0.66$, $BF_{10} = 0.40$)"

Maybe I am missing something, but why use a paired-sample t-test when the upper and lower quartiles constitute different groups of participants? Shouldn't a two-sample t-test be used in this case?

We apologize for the confusion. It is indeed a two-sample t-test.

Along the same lines, I do not understand the link between the number of degrees of freedom reported in the t-test (7) and the number of participants reported in the study (41).

This is also evident when looking at the scatterplot in Figure 3C. How many participants formed the averages and standard errors reported in Figures 3B and 3D? Please clarify.

I have the same comment(s) also for the visual updating task (and related figures), where 13 degrees of freedom are reported in the t-tests. Please clarify.

We thank the reviewer for pointing this out. The number of participants reported in the scatter plots were indeed 42. However, we opted to compare the averages only in the lower and upper quartile of the AQ distribution to avoid dealing with a median split (which would imply a skewed distribution). Of our sample of participants in Exp1, 8 fell into the lower

quartile of the AQ distribution and 8 in the upper quartile (14 deg of freedom); from Exp 2, 8 participants fell in the lower and 7 in the upper (13 deg of freedom).

We now fixed the values accordingly.

Reviewer #3 (Recommendations For The Authors):

(1) The language can be a bit misleading (especially the title and abstract) as it wasn't always clear that the participants don't actually have clinical ASD. I'd suggest avoiding using words like "symptom" as that would indicate clinical severity, and using words like "traits/characteristics" instead for more precise language.

We apologize for the misleading terminology used. Now fixed.

(2) In the Intro: "...perfect compensation results in a vertical trajectory, while small biases indicate stabilization issues23-25." This is a bit confusing without knowing the details of the paradigm. Consider clarifying or at least referring to Figure 4.

Thank you.

(3) In the Results: "This result was also confirmed by a paired sample t-test ($t(7) = 0.66$;..." This is confusing as a two-sample t-test is the appropriate test here. Also, the degree of freedom seems very low - could the authors clarify how many participants are in each subgroup (i.e., low vs. high AQ quartile), for both experiments?

Of our sample of participants in Exp1 8 fell into the lower quartile of the AQ distribution and 8 in the upper quartile (14 deg of freedom); from Exp 2, 8 participants fell in the lower and 7 in the upper (13 deg of freedom).

(4) In the Methods: Experiment 2: "The first dot could appear randomly above or below gaze level at a fixed horizontal location, halfway between the two fixations ($x = 0$, $y = -5\{\text{degree sign}\}$ or $+5\{\text{degree sign}\}$ depending on the trial). The second dot was then shown orthogonal to the first one at a variable horizontal location ($x = 5\{\text{degree sign}\} \{plus minus\} 2.5\{\text{degree sign}\}$)." This would mean that the position of the 2nd dot relative to the 1st one would be $2.5\{\text{degree sign}\} - 7.5\{\text{degree sign}\}$, but the task description in Results and Figure 5A would suggest the horizontal location of the second dot is $x = 0\{\text{degree sign}\} \{plus minus\} 2.5\{\text{degree sign}\}$. Which one is correct?

The second option is the correct one. We now fixed the typo in the Methods part.

(5) There is another study that examined oculomotor efference copies in children with ASD using a similar trans-saccadic perception task (Yao et al., 2021, Journal of Vision). In that study, they found a correlation between task performance and an ASD motor symptom (repetitive behavior). This seems quite relevant to the authors' hypothesis and discussion.

We thank the reviewer for the suggestion. We now added the mentioned paper in the discussion.

(6) Please proofread the entire paper carefully as there were multiple grammatical and spelling errors.

Thank you.

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